Quaternionic Logic: A G-Lattice Unifying Boolean and Fuzzy Frameworks

J. Arturo Ornelas Brand
Independent Researcher
arturoornelas62@gmail.com

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Abstract

We define a formal logic over the four-dimensional space $V = [0,1] \times [-1,1]^3$, where the real component encodes truth and the imaginary components encode epistemic character. Logical connectives (AND, OR, implication) form a bounded residuated lattice, equipped with an involutive negation (NOT). An SU(2) group acts on truth values by quaternionic conjugation, defining context transformations that preserve truth while rotating epistemic character.

The central result is a *commutation theorem*: lattice connectives commute with the group action if and only if they operate exclusively on the real component. Boolean logic therefore emerges as the *context-free* (group-invariant) special case, while the full four-dimensional logic is context-dependent. Boolean and Gödel fuzzy logic are recovered exactly as dimensional restrictions; four additional logics (modal, trivalent, ordinal, probabilistic) admit interpretive correspondences.

The combined structure is a *G-lattice*: a residuated lattice equipped with a non-abelian group action. To our knowledge, no precedent for this specific combination—residuated lattice with SU(2) conjugation and real-part invariance—exists in the literature.

All results are verified computationally at machine precision ($\sim 10^{-16}$) over 50,000 test cases.

1 Introduction

Boolean and fuzzy logics are widely regarded as distinct formal systems. Boolean logic operates on $\{0,1\}$ with classical connectives; fuzzy logic generalises to $[0,1]$ with t-norms [Zadeh, 1965, Hájek, 1998]. Modal, many-valued, and probabilistic logics add further structure. Attempts to unify these frameworks typically embed one system within another or construct meta-logics that contain multiple systems as fragments [Cignoli et al., 2000, Esteva & Godo, 2001].

This paper proposes a different approach. Rather than embedding logics within a common language, we show that they arise as *dimensional restrictions* of a single four-dimensional structure—and that the mechanism distinguishing them is *context sensitivity*.

The construction builds on the quaternionic operator $\mathcal{O} = \{0, 1, +, i, j, k\}$ introduced in the companion paper [Ornelas Brand, 2026], which associates to each concept a four-dimensional coordinate (r, i, j, k) encoding crystallised truth (r), potentiality (i), verification (j), and selection (k).

Contributions.

1. A bounded residuated lattice on $[0,1] \times [-1,1]^3$ that recovers Boolean and Gödel fuzzy logic exactly, with interpretive correspondences to four others (Section 3).
2. A commutation theorem characterising Boolean logic as the group-invariant sublattice (Section 4).

3. A soundness proof for inference rules across all contexts (Section 5).
4. A completeness theorem reducing G-lattice tautologies to $G^{\sim}$ (Gödel logic with involutive negation) via a real-part independence lemma (Section 6).
5. A formal argument that the imaginary dimensions are operationally distinguishable, not decoration (Section 7).
6. Computational verification of all results at machine precision (Section 9).

2 Preliminaries

2.1 Quaternions

The quaternion algebra $\mathbb{H}$ is the four-dimensional real algebra spanned by $\{1, i, j, k\}$ with multiplication rules $i^2 = j^2 = k^2 = ijk = -1$. Every quaternion $q = r + ai + bj + ck$ has real part $\mathrm{Re}(q) = r$ and imaginary part $\mathrm{Im}(q) = (a, b, c)$. The conjugate is $\bar{q} = r - ai - bj - ck$, and for unit quaternions ($|q| = 1$), $q^{-1} = \bar{q}$.

The group SU(2) of unit quaternions acts on $\mathbb{H}$ by conjugation: $q \cdot a = q\, a\, \bar{q}$. This action preserves the real part, the imaginary norm, and the total norm:

$$\mathrm{Re}(q\, a\, \bar{q}) = \mathrm{Re}(a), \tag{1}$$

$$\|\,\mathrm{Im}(q\, a\, \bar{q})\| = \|\,\mathrm{Im}(a)\|, \tag{2}$$

$$\|q\, a\, \bar{q}\| = \|a\|. \tag{3}$$

2.2 Residuated lattices

A *bounded residuated lattice* is a structure $(L, \wedge, \vee, \otimes, \rightarrow, 0, 1)$ where $(L, \wedge, \vee, 0, 1)$ is a bounded lattice, $\otimes$ is a monoidal operation (the t-norm), and $\rightarrow$ is its residuum satisfying $a \otimes b \leq c \iff a \leq b \rightarrow c$ [Galatos et al., 2007].

2.3 The Four Kingdoms

Following Ornelas Brand [2026], each axis of the quaternionic operator corresponds to an irreducible mode of information:

- r: what crystallises, fixes, defines (resolved truth),
- i: what flows, invents, projects (potentiality),
- j: what measures, compares, orders (verification),
- k: what selects, prunes, advances (recursion).

3 The Lattice of Connectives

Definition 1 (Truth values)**.** *A* quaternionic truth value *is an element* $a = (r, i, j, k) \in V = [0, 1] \times [-1, 1]^3$*. The component* $r \in [0, 1]$ *is the* degree of truth*. The triple* $(i, j, k) \in [-1, 1]^3$ *is the* epistemic character*.*

Definition 2 (Connectives)**.** *Define component-wise operations on* $V = [0, 1] \times [-1, 1]^3$*:*

$$\mathrm{AND}(a, b) = (\min(a_r, b_r),\ \min(a_i, b_i),\ \min(a_j, b_j),\ \min(a_k, b_k)), \tag{4}$$

$$\mathrm{OR}(a, b) = (\max(a_r, b_r),\ \max(a_i, b_i),\ \max(a_j, b_j),\ \max(a_k, b_k)), \tag{5}$$

$$\mathrm{NOT}(a) = (1 - a_r,\ -a_i,\ -a_j,\ -a_k), \tag{6}$$

$$(a \rightarrow b)_m = \begin{cases} 1 & \textit{if } a_m \leq b_m, \\ b_m & \textit{otherwise}, \end{cases} \quad \textit{for each component } m \in \{r, i, j, k\}. \tag{7}$$

Proposition 3 (Lattice structure)**.** *$(V, \mathrm{AND}, \mathrm{OR}, \to, \mathbf{0}, \mathbf{1})$ is a bounded residuated lattice with bounds $\mathbf{0} = (0, -1, -1, -1)$ and $\mathbf{1} = (1, 1, 1, 1)$ under the component-wise order.*

Proof. Component-wise min and max on the bounded intervals $[0, 1] \times [-1, 1]^3$ form a bounded distributive lattice with $\mathbf{0} \leq a \leq \mathbf{1}$ for all $a \in V$.

Residuation. We verify $\mathrm{AND}(a, b) \leq c$ iff $a \leq (b \to c)$ for all $a, b, c \in V$, component by component. Fix any component m. ($\Rightarrow$): If $b_m \leq c_m$, then $(b \to c)_m = 1 \geq a_m$. If $b_m > c_m$, then $(b \to c)_m = c_m$ and we need $a_m \leq c_m$. From $\min(a_m, b_m) \leq c_m$ and $b_m > c_m$ it follows that $a_m \leq b_m$, hence $\min(a_m, b_m) = a_m \leq c_m$. ($\Leftarrow$): If $b_m \leq c_m$, then $\min(a_m, b_m) \leq b_m \leq c_m$. If $b_m > c_m$, then $a_m \leq (b \to c)_m = c_m$, so $\min(a_m, b_m) \leq a_m \leq c_m$. □

Remark 4 (Involutive negation)**.** *The negation* (6) *is* not *the lattice negation $a \to \mathbf{0}$ (which is Gödel negation: 1 if $a = \mathbf{0}$, $\mathbf{0}$ otherwise). It is an additional* involutive *negation satisfying* $\mathrm{NOT}(\mathrm{NOT}(a)) = a$ *and the De Morgan laws:* $\mathrm{NOT}(\mathrm{AND}(a, b)) = \mathrm{OR}(\mathrm{NOT}(a), \mathrm{NOT}(b))$*, since* $1 - \min(x, y) = \max(1 - x, 1 - y)$ *and* $-\min(x, y) = \max(-x, -y)$*. The expanded structure is a residuated lattice with involutive negation in the sense of Esteva et al. [2000].*

3.1 Specialisations

Theorem 5 (Exact recovery)**.** *Under dimensional restrictions, the structure* $(V, \mathrm{AND}, \mathrm{OR}, \mathrm{NOT}, \to)$ *reduces exactly to:*

1. ***Boolean**: $i = j = k = 0$, $r \in \{0, 1\} \Rightarrow$ classical propositional logic.*
2. ***Gödel fuzzy**: $i = j = k = 0$, $r \in [0, 1] \Rightarrow$ Gödel fuzzy logic (min t-norm).*

Proof. Under restriction (1), V collapses to $\{0, 1\}$: min is Boolean AND, max is Boolean OR, $1 - r$ is classical negation, and the residuum gives material implication. Under (2), V collapses to $[0, 1]$ with the standard Gödel connectives [Hájek, 1998]. Both are isomorphisms, not merely embeddings. □

Remark 6 (Interpretive correspondences)**.** *Four additional logics admit* interpretive correspondences *(not formal isomorphisms) under dimensional restriction:*

3. ***Modal**: $j = k = 0$, $r, i \in [0, 1]$, reading necessity as $r = i = 1$ and possibility as $r > 0 \vee i > 0$.*
4. ***Trivalent**: $i = k = 0$, $j \in \{-1, 0, +1\}$, encoding the third value via the sign of j.*
5. ***Ordinal**: $j = k = 0$, i encoding rank.*
6. ***Probabilistic**: $j = k = 0$, i encoding variance.*

These are suggestive readings of the restricted subspaces, not claims that a modal or probabilistic logician would recognise their formal system. The value of the quaternionic lattice rests on the exact recovery of Boolean and Gödel logic and the commutation theorem (Theorem 11), not on these looser analogies.

4 The G-Lattice: Context as Group Action

Definition 7 (Context transformation)**.** *A* context *is a unit quaternion $q \in \mathrm{SU}(2)$. The context transformation of a truth value a is the conjugation action:*

$$q \cdot a \;=\; q\, a\, \bar{q}.$$

Definition 8 (G-lattice)**.** *The* quaternionic G-lattice *is the triple $(L, G, \cdot)$ where:*

- *$L = (V, \mathrm{AND}, \mathrm{OR}, \to, \mathbf{0}, \mathbf{1})$ is the residuated lattice from Proposition 3, equipped with the involutive negation* NOT *from Remark 4,*
- *$G = \mathrm{SU}(2)$ is the group of unit quaternions,*
- *$\cdot$ is the conjugation action from Definition 7.*

Proposition 9 (Action properties). *The conjugation action satisfies:*

1. $\mathrm{Re}(q \cdot a) = \mathrm{Re}(a)$ *(truth is objective),*
2. $\|\mathrm{Im}(q \cdot a)\| = \|\mathrm{Im}(a)\|$ *(epistemic weight is conserved),*
3. $\|q \cdot a\| = \|a\|$ *(information is conserved),*
4. $q \cdot (a \cdot_{\mathbb{H}} b) = (q \cdot a) \cdot_{\mathbb{H}} (q \cdot b)$ *(algebra automorphism).*

Proof. Properties (1)–(3) follow from equations (1)–(3). Property (4): $q(ab)\bar{q} = (qa\bar{q})(qb\bar{q})$ by associativity and $\bar{q}q = 1$. □

Remark 10 (Domain of the action). *Conjugation acts on* $\mathrm{Im}(a)$ *as an* $\mathrm{SO}(3)$ *rotation, preserving* $\|\mathrm{Im}\|$ *but potentially redistributing components beyond* $[-1, 1]$ *when* $\|\mathrm{Im}(a)\| > 1$. *The action is therefore defined on the ambient space* $[0, 1] \times \mathbb{R}^3 \supseteq V$. *Every purely real element* $(r, 0, 0, 0)$ *is a fixed point, so the Boolean sublattice (*$\mathrm{Im} = 0$*) is always preserved. The logical content of the G-lattice depends only on* Re *(by Lemma 19), so the domain extension does not affect tautologies or derivability.*

Theorem 11 (Commutation theorem). *Let* $f : V^n \to V$ *be any connective (AND, OR, NOT,* $\to$*, or a composition thereof). Then*

$$q \cdot f(a_1, \ldots, a_n) = f(q \cdot a_1, \ldots, q \cdot a_n) \quad \textit{for all } q \in \mathrm{SU}(2)$$

if and only if f *depends only on the real components* $(\mathrm{Re}(a_1), \ldots, \mathrm{Re}(a_n))$.

Proof. ($\Leftarrow$) If f depends only on real parts, then $\mathrm{Re}(q \cdot a_m) = \mathrm{Re}(a_m)$ by Proposition 9(1), so f receives the same inputs regardless of q. The output has zero imaginary part (since f outputs only a real-part result), and conjugation fixes real quaternions. Hence both sides are equal.

($\Rightarrow$) Suppose f depends on some imaginary component. Choose a such that the imaginary part of $f(a)$ is nonzero. Then there exists $q \in \mathrm{SU}(2)$ that rotates $\mathrm{Im}(f(a))$ to a different direction, while $f(q \cdot a)$ computes min/max on the rotated components of a—which generically differs from the rotation of min/max of the original components (since min does not commute with rotation). Hence the two sides differ. □

Example 12 (Explicit non-commutation). *Let* $a = (0.7, 0.3, 0.5, 0.1)$, $b = (0.4, 0.8, 0.2, 0.6)$, *and* $q = \frac{1}{\sqrt{2}}(1, 1, 0, 0)$. *Then* $q \cdot \mathrm{AND}(a, b) = (0.4,\ 0.3,\ -0.1,\ 0.2)$ *while* $\mathrm{AND}(q \cdot a,\ q \cdot b) = (0.4,\ 0.3,\ -0.6,\ 0.2)$*; the* j*-components differ (*-0.1 *vs* -0.6*). The real parts agree (*$0.4 = 0.4$*), confirming that the non-commutation is purely epistemic.*

Corollary 13 (Boolean logic is context-free). *The sublattice* $\{a \in V : \mathrm{Im}(a) = 0\}$ *is the unique maximal sublattice on which all connectives commute with every context transformation. This is the Boolean/fuzzy restriction.*

Definition 14 (Context sensitivity). *The* context sensitivity *of a truth value* a *is*

$$\sigma(a) = 2\,\|\mathrm{Im}(a)\|.$$

This equals the maximum displacement $\max_{q \in \mathrm{SU}(2)} \|q \cdot a - a\|$.

Remark 15. *The context sensitivity provides a continuous spectrum:* $\sigma = 0$ *corresponds to Boolean logic (context-free), and increasing* σ *gives progressively more context-dependent logics. For truth values in* V*, the spectrum is parameterised by* $\|\mathrm{Im}(a)\| \in [0, \sqrt{3}]$.

5 Soundness

Definition 16 (Truth predicate). *For a threshold $\tau \in [0,1]$, a truth value a is* true *if* $\mathrm{Re}(a) \geq \tau$.

The inference rules of the quaternionic G-lattice are:

R1. **Modus ponens:** From a and $a \to b$, infer b.
R2. **Context shift:** From a, infer $q \cdot a$ for any $q \in \mathrm{SU}(2)$.
R3. **Context composition:** $q_2 \cdot (q_1 \cdot a) = (q_2 q_1) \cdot a$.

Theorem 17 (Soundness). *The inference rules **R1–R3** are sound with respect to the truth predicate of Definition 16.*

Proof. **R1** (Modus ponens): If $\mathrm{Re}(a) \geq \tau$ and $\mathrm{Re}(a \to b) \geq \tau$, then either $a_r \leq b_r$ (in which case $b_r \geq a_r \geq \tau$) or $a_r > b_r$ (in which case $\mathrm{Re}(a \to b) = b_r \geq \tau$). In both cases $\mathrm{Re}(b) \geq \tau$. This is the standard soundness argument for the Gödel residuum [Hájek, 1998, Galatos et al., 2007].

R2 (Context shift): By Proposition 9(1), $\mathrm{Re}(q \cdot a) = \mathrm{Re}(a) \geq \tau$.

R3 (Context composition): By associativity of quaternion multiplication: $q_2(q_1 a \bar{q}_1)\bar{q}_2 = (q_2 q_1) a (\overline{q_2 q_1})$. □

Remark 18. *Soundness decomposes cleanly: **R1** is inherited from the residuated lattice literature; **R2** follows from the real-part invariance of conjugation; **R3** is the group law. No new proof technique is required—the G-lattice soundness is the* product *of two well-understood soundness results.*

6 Completeness

Lemma 19 (Real-part independence). *For every propositional formula φ built from AND, OR, NOT, $\to$ and every valuation $v : \mathrm{Atoms} \to V$,*

$$\mathrm{Re}([\![\varphi]\!]_v) \text{ depends only on } (\mathrm{Re}(v(p_1)), \ldots, \mathrm{Re}(v(p_n))).$$

That is, the imaginary components of the atoms do not affect the real part of the result.

Proof. By structural induction on φ. For each connective:

$$\begin{aligned}
\mathrm{Re}(\mathrm{AND}(a,b)) &= \min(\mathrm{Re}(a), \mathrm{Re}(b)),\\
\mathrm{Re}(\mathrm{OR}(a,b)) &= \max(\mathrm{Re}(a), \mathrm{Re}(b)),\\
\mathrm{Re}(\mathrm{NOT}(a)) &= 1 - \mathrm{Re}(a),\\
\mathrm{Re}(a \to b) &= \begin{cases} 1 & \text{if } \mathrm{Re}(a) \leq \mathrm{Re}(b),\\ \mathrm{Re}(b) & \text{otherwise.}\end{cases}
\end{aligned}$$

In each case the output's real part is a function of the inputs' real parts alone. The induction step follows by compositionality. □

Theorem 20 (Completeness). *Let φ be a propositional formula built from* $\mathrm{AND}, \mathrm{OR}, \mathrm{NOT}, \to$.

1. *φ is a G-lattice tautology ($\mathrm{Re}([\![\varphi]\!]_v) \geq \tau$ for every $v : \mathrm{Atoms} \to V$) if and only if φ is a tautology of the $[0,1]$-algebra $(\min, \max, 1-x, \to_G)$, where $\to_G$ is the Gödel residuum.*
2. *For the negation-free fragment (φ built from $\mathrm{AND}, \mathrm{OR}, \to$ only), this reduces to Gödel–Dummett completeness [Dummett, 1959, Hájek, 1998].*
3. *For the full fragment (with* NOT*), the logic is $G^{\sim}$—Gödel logic expanded with an involutive negation—which is complete by Esteva et al. [2000].*

Proof. By Lemma 19, $\mathrm{Re}([\![\varphi]\!]_v)$ depends only on the real parts of the atomic valuations. Therefore φ is a G-lattice tautology iff it is a tautology of the real-part projection, which is the $[0,1]$-algebra with min, max, the Gödel residuum, and the involutive negation $1-x$.

For the negation-free fragment, this algebra is the standard Gödel chain, and completeness follows from Dummett [1959], Hájek [1998]. For the full fragment, the algebra is a Gödel algebra expanded with involutive negation; completeness of this class ($G^{\sim}$) is established in Esteva et al. [2000]. □

Remark 21. *The involutive negation* $\mathrm{NOT}(a) = 1 - a$ *is* not *the Gödel negation* $a \to \mathbf{0}$. *In particular,* $\mathrm{NOT}(\mathrm{NOT}(a)) = a$ *is a tautology (double negation holds), while* $(\varphi \to \psi) \to (\mathrm{NOT}(\psi) \to \mathrm{NOT}(\varphi))$ *(contraposition) is not. These are characteristic of* $G^{\sim}$*, not of standard Gödel logic.*

The imaginary dimensions enrich the logic with context sensitivity, epistemic character, and path-dependent reasoning, but they do not alter which formulas are tautologies. The "new" content of the quaternionic G-lattice is not in its tautologies but in the structure around *them: the group action, the context spectrum, and the cause–effect chains.*

7 Epistemic Dimensions Are Not Decoration

A potential objection is that since $\mathrm{Re}(a)$ is always context-invariant, the imaginary dimensions (i, j, k) are "decoration" that do not affect logical reasoning. We present three arguments against this objection.

7.1 Operational distinguishability

Definition 22 (Decision function)**.** *A* decision function *is any map* $D : V \to \mathcal{A}$ *from truth values to an action space* $\mathcal{A}$ *that depends on the full quaternionic truth value, not only on* $\mathrm{Re}(a)$.

Proposition 23. *If* D *depends on* $\mathrm{Im}(a)$*, then there exist truth values* a, b *with* $\mathrm{Re}(a) = \mathrm{Re}(b)$ *such that* $D(a) \neq D(b)$.

Proof. Take $a = (r, \alpha, 0, 0)$ and $b = (r, 0, \alpha, 0)$ for any $r \in (0,1)$ and $\alpha \neq 0$. Then $\mathrm{Re}(a) = \mathrm{Re}(b) = r$ but $\mathrm{Im}(a) \neq \mathrm{Im}(b)$, and by assumption D distinguishes them. □

Example 24 (The anchor argument)**.** *A goal is scored:* $a = (0.9, 0.6, 0.7, 0.5)$*. Both teams agree on the fact (*$\mathrm{Re} = 0.9$*). Let* q_{red} *be a context rotation (a different "anchor"):*

$$q_{\mathrm{red}} \cdot a = (0.9,\ 0.6,\ -0.7,\ -0.5).$$

The truth is identical. The epistemic character differs: one team reads positive verification ($+j$*) and celebrates; the other reads negative verification (*$-j$*) and laments. The decision (celebrate or lament) depends on* (i, j, k)*, not only on* r.

7.2 The panorama

Averaging over all contexts yields a pure-real result:

$$\int_{\mathrm{SU}(2)} q \cdot a \, d\mu(q) = \big(\mathrm{Re}(a),\ 0,\ 0,\ 0\big),$$

where μ is the Haar measure on SU(2). This is the "view from nowhere"—Boolean logic. Agents, however, do not live in the panorama; they are situated in specific contexts. The imaginary dimensions encode the *situated perspective* that determines behaviour.

7.3 Epistemic quality

Two propositions with the same truth ($\mathrm{Re} = 0.7$) but different epistemic profiles carry different information:

- (0.7, 0.8, 0.1, 0.2): high potentiality, low verification—a *hypothesis*.
- (0.7, 0.1, 0.8, 0.2): low potentiality, high verification—an *observation*.

A rational agent treats hypotheses and observations differently, even at equal truth values. The dimensions encode *epistemic quality*, which is decision-relevant.

8 Cause–Effect Chains

Since SU(2) is non-abelian, the composition of context changes is order-dependent:

$$q_2 \cdot (q_1 \cdot a) \neq q_1 \cdot (q_2 \cdot a) \quad \text{in general.}$$

All orderings preserve the same $\mathrm{Re}(a)$, but produce different epistemic signatures Im. The process retains *memory of the path taken*: the order of perspective changes leaves a distinct epistemic trace.

Proposition 25 (Path dependence)**.** *For n distinct context changes $q_1, \ldots, q_n \in \mathrm{SU}(2)$, the $n!$ orderings of application to a truth value a all yield the same $\mathrm{Re}(a)$ but generically yield $n!$ distinct imaginary parts.*

Proof. Re is invariant under every conjugation (iterated application of Proposition 9(1)). For the imaginary part, distinct orderings correspond to distinct elements of SU(2) (since the group is non-abelian), which generically produce distinct rotations of $\mathrm{Im}(a)$. □

This formalises the notion that "imagining then verifying" and "verifying then imagining" leave different epistemic signatures, even when they agree on the truth. The non-commutativity is the algebraic encoding of irreversible cause–effect chains.

9 Computational Verification

All results were verified computationally using NumPy in Python. Random test cases were drawn uniformly from the appropriate domains.

Table 1: Summary of computational verification.

Test	**N**	**Violations**	**Max error**
Specialisations (2 exact + 4 interpretive)	—	0	exact
De Morgan laws (extended domain)	50 000	0	0
Re invariance under conjugation	50 000	0	5.55×10^{-16}
$\lVert \mathrm{Im} \rVert$ invariance under conjugation	10 000	0	6.66×10^{-16}
Algebra automorphism	10 000	0	$< 10^{-12}$
Commutation (Boolean AND)	10 000	0	3.33×10^{-16}
Non-commutation (full AND)	10 000	9 634	1.16
Non-commutation (full OR)	10 000	9 637	1.23
Non-commutation (NOT)	10 000	10 000	2.73
Modus ponens (all contexts)	50 000	0	exact
Context shift preserves truth	50 000	0	exact
Re of compound formula invariant	50 000	0	9.99×10^{-16}
Domain closure (r in $[0, 1]$)	50 000	0	exact
Decision distinguishability	50 000	—	49.6% changed
Panorama ($\mathrm{Im} \to 0$)	10 000	—	$\lVert \mathrm{Im} \rVert < 0.005$
Re-independence (all connectives)	100 000	0	0
Re-independence (7 compounds)	100 000	0	0
$G^{\sim}$ tautologies (6 formulas)	100 000	0	confirmed
Non-tautologies (4 formulas)	100 000	0	confirmed

Source code is available at `scripts/{connectives, context_bridge, verify_resolutions, verify_completeness}.py` in the companion repository.

10 Related Work

Table 2: Comparison with existing algebraic structures for logic.

Structure	**Source**	**Relation**
Residuated lattice	Galatos et al. [2007]	Layer 1 of our structure
BL-algebra	Hájek [1998]	Our lattice under fuzzy restriction
MV-algebra	Cignoli et al. [2000]	Related to Łukasiewicz specialisation
Quaternionic fuzzy sets	Dai [2023]	Quaternion-valued membership; no group action, no logic
Quantale	Mulvey [1986]	Complete lattice + monoid; ours uses non-abelian group
Quantum logic	Birkhoff & von Neumann [1936]	Projections on Hilbert spaces, not t-norms
Substructural logic	Restall [2000]	Non-commutative connectives, no group action

The closest antecedent is Dai [2023], who defines quaternionic fuzzy *sets* with quaternion-valued membership functions bounded by unit modulus. The present work differs in three respects: (i) we define a propositional *logic* with inference rules, soundness, and completeness, not only set operations; (ii) we add a group action that separates truth from epistemic character via the commutation theorem; (iii) our domain is component-wise bounded ($[0, 1] \times [-1, 1]^3$) rather than modulus-bounded.

More broadly, the distinguishing feature of the present work is that the group action is not applied to the connectives (which remain commutative) but to the *truth values themselves*, with a clean invariance theorem separating objective truth from perspectival epistemic character.

11 Conclusion

We have shown that the quaternionic operator from Ornelas Brand [2026] supports a G-lattice that genuinely unifies Boolean and fuzzy frameworks. The unification is not mere containment: it is mediated by a commutation theorem that characterises Boolean logic as the context-free special case. The four-dimensional logic introduces context sensitivity, path-dependent epistemic chains, and operationally distinguishable epistemic dimensions—properties absent from any individual restriction.

Completeness. The G-lattice inherits completeness from $G^{\sim}$ (Gödel logic with involutive negation) via the real-part independence lemma (Theorem 20). The negation-free fragment reduces to standard Gödel–Dummett. The imaginary dimensions do not alter tautologies—they enrich the surrounding structure.

Future work. Empirical validation on the 72 semantic primitives from Ornelas Brand [2026] is planned.

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